Gradient descent in weight space is kernel descent in activity space

GitHub · Interactive Demo (JSX)

Consider a feedforward network with weights $\mathbf{W}$ and neural activities $A_i(\mathbf{W})$, trained to minimise a loss L. Gradient descent updates every weight simultaneously:

$$\Delta W_\alpha = -\eta \, \frac{\partial L}{\partial W_\alpha}\,, \qquad \forall\, \alpha. \tag{1}$$

What is the induced change in the activity of neuron i? Because A_i is a deterministic function of all weights, we can apply the chain rule over the full parameter vector:

$$\Delta A_i \;=\; \sum_\alpha \frac{\partial A_i}{\partial W_\alpha}\, \Delta W_\alpha \;=\; -\eta \sum_\alpha \frac{\partial A_i}{\partial W_\alpha}\, \frac{\partial L}{\partial W_\alpha}\,. \tag{2}$$

Expanding $\partial L/\partial W_\alpha = \sum_k (\partial L/\partial A_k)\,(\partial A_k/\partial W_\alpha)$ and exchanging the sums yields

$$\boxed{\Delta A_i \;=\; -\eta \sum_k \Theta_{ik}\, \frac{\partial L}{\partial A_k}\,,} \tag{3}$$

where

$$\Theta_{ik} \;\coloneqq\; \sum_\alpha \frac{\partial A_i}{\partial W_\alpha}\, \frac{\partial A_k}{\partial W_\alpha} \;=\; \nabla_{\mathbf{W}} A_i \cdot \nabla_{\mathbf{W}} A_k \tag{4}$$

is a neural-tangent-kernel-style Gram matrix defined on internal activities rather than network outputs. In vector form, $\Delta \mathbf{A} = -\eta\, \Theta\, \nabla_{\mathbf{A}} L$: gradient descent in weight space induces *kernel descent* in activity space, with Θ encoding how strongly every pair of neurons shares upstream parameters.

Self term. The diagonal element $\Theta_{ii} = \|\nabla_{\mathbf{W}} A_i\|^2$ collects contributions from *all* weights that influence neuron i—both its own incoming weights and every weight in earlier layers that lies on a path to it. Because $\Theta_{ii} > 0$, the self term is always aligned with $-\partial L/\partial A_i$: each neuron's activity moves down its own loss gradient, with an effective learning rate $\eta\, \Theta_{ii}$ that grows with the depth and width of its input pathways.

Cross terms. The off-diagonal elements $\Theta_{ik} = \nabla_{\mathbf{W}} A_i \cdot \nabla_{\mathbf{W}} A_k$ for $k \neq i$ couple neuron i to every other neuron's loss gradient. Their magnitude reflects parameter sharing: two neurons fed by overlapping early-layer weights will have large $|\Theta_{ik}|$; neurons in disjoint pathways will have $\Theta_{ik} \approx 0$. These cross terms have no reason to be aligned with $\partial L/\partial A_i$ and constitute interference—the side-effect of the rest of the network learning simultaneously.

When is Θ approximately diagonal? The cross terms vanish when $\nabla_{\mathbf{W}} A_i \cdot \nabla_{\mathbf{W}} A_k \approx 0$ for $i \neq k$—that is, when different neurons' activities depend on nearly orthogonal directions in weight space. Several settings push the kernel toward this regime.

Width. At random initialization, two neurons in the same layer share no incoming weights, so their activity gradients are independent random vectors in parameter space. By concentration of measure, their inner product scales as $O(1/\sqrt{d})$ while the diagonal scales as $O(1)$, making Θ increasingly diagonal-dominant as width grows—the internal-activity analogue of the NTK becoming deterministic at infinite width.

Depth and weight sharing. Neurons in the same layer share all earlier-layer weights, which introduces a common component into their parameter gradients. Each neuron's own incoming weights act as independent random projections that decorrelate this shared component in the wide limit. Deeper networks have more shared parameters but also more such projections; in practice, the cross terms remain significant at finite width.

Learning dynamics. Gradient descent itself may reinforce diagonal dominance. Training is known to decorrelate neural representations as the network learns to tile the input space efficiently. To the extent that decorrelated activities imply decorrelated parameter gradients—since $\partial A_i/\partial W_\alpha$ depends on upstream activations feeding into i—learning should progressively shrink the off-diagonal entries of Θ. The link between activity and gradient decorrelation is heuristic and deserves formal treatment, but it suggests that trained networks may be closer to the diagonal regime than their initializations.

Upshot. The activity change ΔA_i is a weighted combination of the loss gradients of *all* neurons, with weights given by the kernel Θ. When the kernel is diagonally dominant ($\Theta_{ii} \gg \sum_{k\neq i} |\Theta_{ik}|$), activity dynamics closely track the negative loss gradient, $\Delta A_i \approx -\eta\, \Theta_{ii}\, \partial L/\partial A_i$. In this regime, each neuron's activity change is simply proportional to what backpropagation says it should be—an approximation that improves with width and, arguably, over the course of training.

Caveat: the reverse direction. The converse is weaker: observing that activities follow their loss gradient constrains the projection of $\Delta \mathbf{W}$ onto the row space of the Jacobian $\partial A_i/\partial W_\alpha$, but says nothing about the (typically vast) null space of weight changes that leave activities unchanged on the current input. Activity-level gradient descent is therefore consistent with many different weight-level learning rules, of which backpropagation is the minimum-norm solution.

Experimental prediction. If a biological neural network performs any form of gradient descent on its synaptic weights—even an approximate or noisy variant—and is sufficiently wide that Θ is diagonal-dominant, then the observable activity changes must satisfy $\Delta A_i \approx -\eta\,\Theta_{ii}\,\partial L/\partial A_i$. The per-neuron scaling Θ_{ii} is unknown and may vary across neurons, but the *direction* of $\Delta\mathbf{A}$ is pinned: it must be approximately proportional to $-\nabla_{\mathbf{A}} L$. This is directly testable. One can record neural activities across learning trials, estimate $\partial L/\partial A_i$ from a task-fitted model, and check whether the measured ΔA_i is proportional to the predicted gradient—only the direction matters, not the magnitude. The key value of this result is that it converts an untestable claim about synaptic learning rules (which cannot be measured at scale in vivo) into a testable prediction about activity changes (which can be). A failure of proportionality would rule out not just backpropagation but *any* weight-space gradient rule in a network wide enough for diagonal dominance.

Methods. Figures 1–2 numerically verify Eqs. 2–5 in a 3-hidden-layer ReLU MLP trained with online SGD ($\eta = 0.005$) on a 2-class spiral dataset (20 points per class, input dimension 2, output dimension 2). At regular intervals during training, we compute the full activity Jacobian $\partial A_i/\partial W_\alpha$ and the kernel Θ_{ik}, then compare the predicted activity change ΔA from the chain-rule identity (Eq. 2), the kernel equation (Eq. 3), and the diagonal approximation (Eq. 5) against the actual change measured after a single gradient step. Figure 1 shows detailed diagnostics at widths 8 and 48; Figure 2 sweeps width from 4 to 128 (3 seeds each) to quantify how the diagonal approximation improves with width.

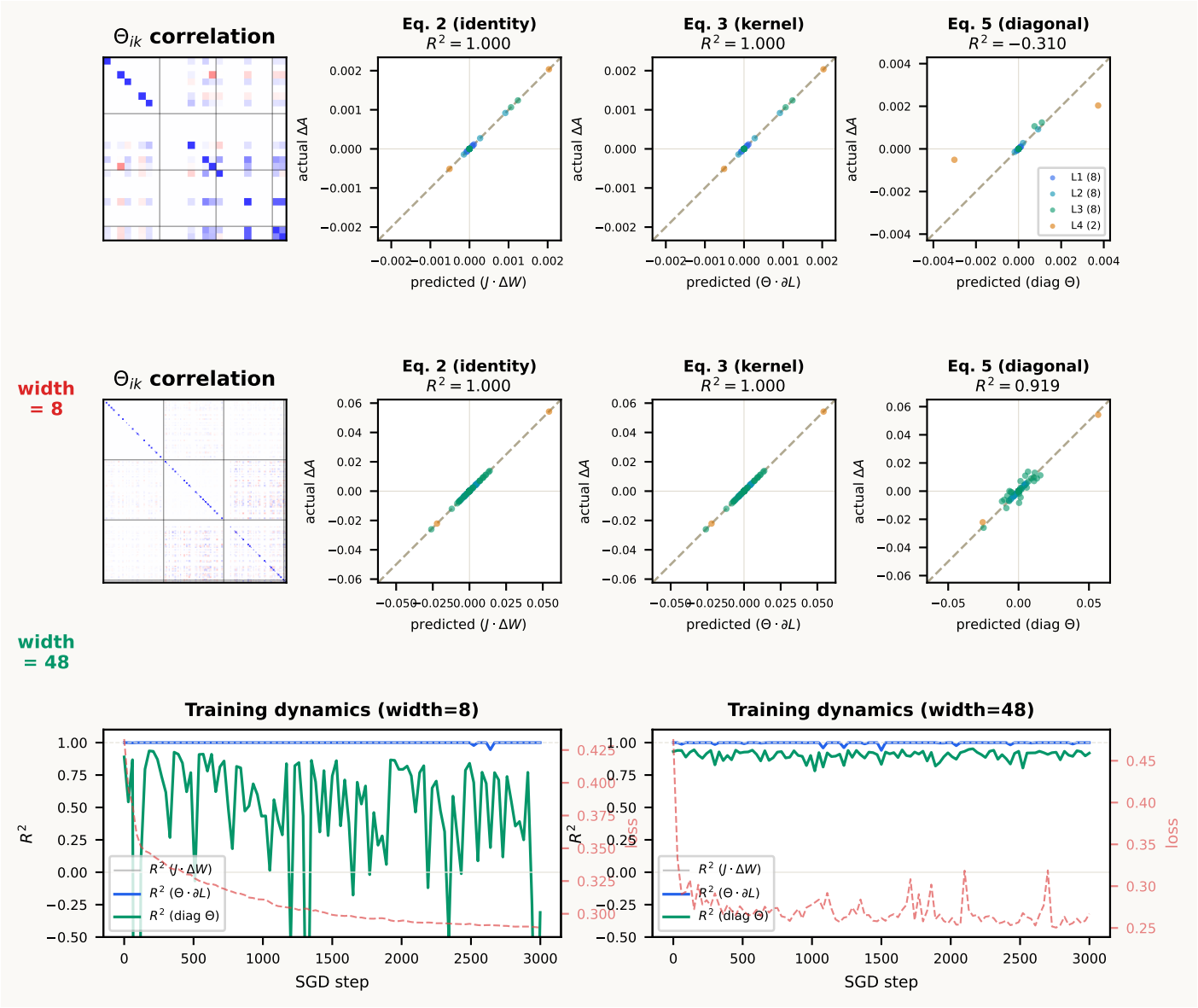

Figure 1: Numerical verification of activity-space kernel descent on a 2-class spiral task (ReLU MLP, 3 hidden layers, online SGD). Top row: width = 8. Middle row: width = 48. Left: normalised Θ_{ik} heatmaps. Centre-left to centre-right: predicted vs. actual ΔA for the chain-rule identity (Eq. 2), the full kernel equation (Eq. 3), and the diagonal approximation (Eq. 5). Bottom row: R^2 and loss over training.

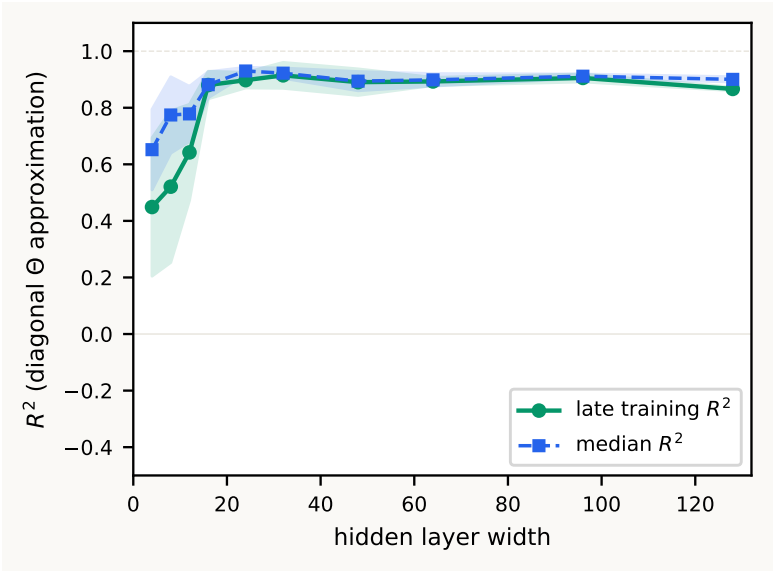

Figure 2: Quality of the diagonal approximation (Eq. 5) as a function of hidden-layer width. Each network has 3 hidden layers trained for 2000 SGD steps; shaded regions show ± 1 s.d. across 3 random seeds. The diagonal approximation is poor at small width ($R^2 < 0$ at width $= 4$) but approaches $R^2 \approx 1$ as width grows, consistent with the off-diagonal entries of Θ becoming negligible in the wide limit.